Problem Set 2

ILA 386 · Data Science for Linguists

Description

For this exercise, you will work with a single data file. The data come from another study we ran in the Speech Learning Lab where we tested participants' ability to identify lexical stress differences such as *canto* ('I sing') and *cantó* ('He/She sang'). Learners completed four tasks:

1) an AX discrimination task (AX), where they heard two stimuli and had to indicate if the two sounds were the same or different (if X was the same as A or different from A);

2) an identification task using orthographic words (ID orthographic), where they heard a single stimulus and had to map it to its written form (*cantó* or *cantó*);

3) an identification task using subject pronouns (ID pronouns), where they heard a single stimulus and had to map it to the correct pronoun (*yo* or *él*);

4) the Spanish LexTALE, a lexical decision task where learners are asked to indicate if the items they see are words or nonwords; this task functions as a proficiency test given that the words are of varying frequency.

Participants completed tasks in the following order: listening tests (AX, ID orthographic, ID pronouns), LexTALE, background questionnaire (not included here). The order in which the listening tests were completed was randomized across participants, and trials within tests were also presented in a random order.

We are interested in the following research questions:

1) How do participants perform on each of these tasks?

2) Do we see differences in performance across the tasks?

3) What is the relationship between proficiency and performance across the tasks?

Wrangling and Cleaning

As you did for PS1, you will need to decide which columns and rows containing meaningful information and should be kept, and which you can filter out. In this file, there is a column indicating whether the participant responded correctly. However, please note that timeouts are coded as incorrect. You need to decide what to do about timeouts.

Engage in whatever data wrangling and cleaning procedures you believe appropriate for the data. Make clear annotations in your code about what you are doing and why.

Data Visualization

Make two descriptive plots: one showing performance on each task (research questions 1 and 2) and one showing the relationship between proficiency and performance across the tasks (research question 3).

Data Analysis: Linear Model (PS1)

Fit a linear model to mean score, including proficiency, task, and proficiency * task as predictors. Check model assumptions, interpret the model output in relation to the research questions, and write up a paragraph describing the results. This should feel familiar to you based on what you did for PS1.

Note: For this analysis, you will need to create a summary data set where each participant has one score per task.

Data Analysis: Logistic Mixed-Effects Model (New)

Fit a logistic model to the trial level (0/1, incorrect/correct) data, including the same predictors. Check model assumptions, interpret the model output in relation to the research questions, and write up a paragraph describing the results.

Comparison

Write a paragraph describing differences between the two approaches. What conclusions do you reach under each approach? What is the same or different?

AI Use

You may not upload this assignment and the associated data sets to an AI platform. You may prompt AI for help, with the following conditions: (1) beforehand, you should make an earnest effort to (re)solve data difficulties yourself (e.g., spend 10-15 minutes thinking through what might be necessary, exploring potential functions in R using '?', reviewing class notes and Winter's book); (2) you must transparently disclose AI use in your code.

Collaboration

You may collaborate with classmates so long as you work independently and come together to review and troubleshoot code. In other words, please do not subdivide this assignment in stages and copy code instead of making an attempt to author code yourself.

Submission

Upload a .Rmd (R markdown) file with complete, annotated code to your repo by the deadline. Your code should be well organized, reflecting the steps described in this assignment.